

How To Take Your IIoT DEVICE From PROTOTYPE TO PRODUCT



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What goes into making an Internet of Things (IIoT) product?

A product development cycle starts with an idea, which, if found to be technically feasible, sees funds raised to start designing the product's prototype. This prototype is modified till it becomes a market-ready product.

For any new product, there should be a focus on achieving quick time-to-market and small production batch size. Let us take a look at how to go about doing this.

The product design flow

There are a few steps that are helpful to those designing products from scratch. These steps are collectively known as the product design flow. It involves all steps in the design process, from start to finish.

Design specifications. The very first step in the product design flow is to define

the specifications of the product. Thus, appearance of the product is finalised, and components are selected. A costing exercise is also included in this step to make sure that the product does not become too expensive. Achieving a balance between features and cost is very important at this stage.

Product costing process is a very important part of the design specification stage. There are some basic rules to be followed for this process. First, a base price is calculated. It is the sum of bill of materials (at minimum order quantity), assembly cost and amortised development cost.

Selling price is decided based on the company's sales method. If it is a traditional sales method, with shop space, sales people and print advertisements, then the base price is multiplied by a factor of 2.7 to determine the sales price of the product. This is the minimum sales price of the

product, and will result in a margin of about 10 to 12 per cent, after tax is deducted.

e-Commerce companies do not have to rent shop space and generally have fewer sales people, so their expenses are usually less as compared to companies with traditional sales methods. For an e-commerce company, base price of the product can be multiplied by a factor of 2.2 to obtain the sales price.

When it comes to increasing margins, companies can

		Penetration of water									
		Non-protected	Protected against vertically falling water drops	Protected against projected water drops when device is tilted up to 15°	Water sprayed at an angle up to 60° from the vertical shall have no harmful effects	Water splashed from any direction shall have no harmful effects	Water projected in jets from any direction shall have no harmful effects	Water projected in powerful jets from any direction shall have no harmful effects	Protected against temporary immersion	Protected against continuous immersion	
Protection against human access to hazardous parts	Protection of equipment against penetration of solid foreign objects		IP x0	IP x1	IP x2	IP x3	IP x4	IP x5	IP x6	IP x7	IP x8
Non-protected	Non-protected	IP 0x	IP 00								
Protected against access to hazardous parts with the back of the hand	Protected against solid foreign objects larger in diameter than 50mm	IP 1x	IP 10	IP 11	IP 12						
Protected against access to hazardous parts with a finger	Protected against solid foreign object larger in diameter than 12.5mm	IP 2x	IP 20	IP 21	IP 22	IP 23					
Protected against access to hazardous parts with a tool larger in diameter than 2.5mm	Protected against solid foreign objects larger in diameter than 2.5mm	IP 3x	IP 30	IP 31	IP 32	IP 33	IP 34				
Protected against access to hazardous parts with a wire larger in diameter than 1.0mm	Protected against solid foreign objects larger in diameter than 1mm	IP 4x	IP 40	IP 41	IP 42	IP 43	IP 44				
Protected against access to hazardous parts with a wire larger in diameter than 1.0mm	Prevents penetration of dust sufficient to cause damage inside the equipment	IP 5x	IP 50			IP 54	IP 55				
Protected against access to hazardous parts with a wire larger in diameter than 1.0mm	Dust-proof	IP 6x	IP 60					IP 65	IP 66	IP 67	IP 68

Fig. 1: International Protection Standards chart